Super-Consistent Cointegration Estimators and Structural Stability of Capital Composition: An Empirical Audit using `cointReg` and `strucchange` for the United States (1925–2021) and Chile (1920–2024)

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Abstract

This paper provides a single-equation empirical audit of the long-run cointegrating relationship between Nonresidential Structures ($K^{\text{NRC}}/K^{\text{NR}}$) and Machinery and Equipment (K^{ME}) across century-long macroeconomic samples in the United States (1925–2021) and Chile (1920–2024). We compare three super-consistent modified OLS estimators from the R `cointReg` package—Fully Modified OLS (FM-OLS; Phillips & Hansen 1990), Dynamic OLS (DOLS; Stock & Watson 1993), and Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014)—and evaluate parameter stability using empirical fluctuation processes (OLS-CUSUM) and Bai-Perron (1998, 2003) dynamic programming break tests from the `strucchange` package. Normalizing the cointegrating vector with Nonresidential Structures on the left-hand side ($k^{\text{NRC}} = \alpha + \gamma k^{\text{ME}} + u_t$), we establish three central findings. First, FM-OLS, DOLS, and IM-OLS yield strikingly consistent $\hat{\gamma}$ estimates across all historical windows, demonstrating that single-equation robustness does not depend on kernel selection or lag specification choices. Second, sub-period estimation reveals strong regime heterogeneity: U.S. Fordist accumulation (1947–1973) exhibits high structural elasticity ($\hat{\gamma} \approx 4.06$–$4.34$), which collapses to statistical instability ($\hat{\gamma} \approx -0.60$, $p > 0.25$) during Neoliberal Restructuring (1974–2008). Third, Bai-Perron endogenous break tests identify major structural shifts at **1947**, **1969**, and **2002** for the U.S. (SupF $= 548.67, p < 0.0001$) and **1932**, **1947**, and **1984** for Chile (SupF $= 378.49, p < 0.0001$). The endogenously determined 1969 U.S. break aligns precisely with the non-parametric LOESS 2nd derivative zero-crossing at 1968, providing independent econometric confirmation of the late-Fordist profit squeeze.

Finally, we reconcile these single-equation modified OLS estimates with Johansen VECM system results. While single-equation models capture the un-augmented levels trajectory, VECM system models with drift controls partial out deterministic trend components, isolating the unit baseline ($\hat{\gamma} \approx 1.0$) of proportional Fordist accumulation. Both approaches decisively confirm that the capital composition state $\tau_t = \ln K^{\text{ME}} - \ln K^{\text{NRC}}$ is the sole econometrically admissible proxy carrying mechanization intensity into the downstream θ_t estimation stage.

1 Introduction: Single-Equation Cointegration vs. System VECMs

Capital accumulation is structurally heterogeneous. Nonresidential structures (K^{NRC} for the U.S.; K^{NR} for Chile) form the physical envelope of capacity expansion, whereas machinery and equipment (K^{ME}) embodies the intensive choice of technique through which capitalists respond to distributional pressure at the shop floor. The long-run cointegrating regression ($k_t^{\text{NRC}} = \alpha + \gamma k_t^{\text{ME}} + u_t$) frames the elasticity $\gamma = \frac{\partial k^{\text{NRC}}}{\partial k^{\text{ME}}}$ as the structural envelope response to machinery deepening.

In standard time series econometrics, static OLS estimation of cointegrated $I(1)$ series yields super-consistent estimates ($\hat{\beta}_{\text{OLS}} - \beta = O_p(T^{-1})$; Stock 1987). However, static OLS produces non-Gaussian limiting distributions contaminated by two nuisance parameter sources: (1) *long-run endogeneity* ($\Omega_{12} \neq 0$) arising from feedback between cointegration residuals and regressor innovations, and (2) *second-order serial correlation bias* ($\Delta_{21} \neq 0$) in the residual process (Phillips & Hansen 1990). These asymptotic distortions render standard t-statistics and hypothesis tests invalid.

This paper evaluates three single-equation modified OLS estimators that purge these asymptotic biases:

- **Fully Modified OLS (FM-OLS; Phillips & Hansen 1990)**: A non-parametric kernel approach that transforms the dependent variable and applies an additive bias correction vector using long-run covariance matrices $(\hat{\Omega}, \hat{\Delta})$.
- **Dynamic OLS (DOLS; Stock & Watson 1993)**: A parametric approach that augments the static regression with leads and lags of first-differenced regressors ($\Delta k^{\text{ME}}_{t+j}$) to absorb endogeneity and short-run dynamics.
- **Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014)**: A partial-sum integration transformation that eliminates kernel bandwidth and lead/lag selection choices while employing Fixed-b asymptotic theory for finite-sample inference.

We evaluate these estimators across five historical periodization windows per country defined strictly by non-parametric LOESS 2nd derivative zero-crossings (1933, 1968, 2008 for the U.S.; 1936, 1971, 1982 for Chile). In addition, we execute formal structural stability tests using the R `strucchange` package (Zeileis et al. 2002), employing empirical fluctuation processes (OLS-CUSUM) and Bai-Perron (1998, 2003) dynamic programming break tests.

Transition to Section 2: To establish the theoretical properties of these estimators, Section 2 details the mathematical resolution of second-order bias and endogeneity in single-equation cointegration.

2 Econometric Foundations of Super-Consistent Modified OLS

Consider the cointegrated regression model between $I(1)$ variables $y_t = k^{\text{NRC}}_t$ and $x_t = k^{\text{ME}}_t$:

$$y_t = x_t\gamma + u_{1t}, \quad \Delta x_t = u_{2t}, \tag{1}$$

where $u_t = (u_{1t}, u_{2t})'$ is a mean-zero stationary vector error process with long-run covariance matrix Ω and total one-sided covariance matrix Δ:

$$\Omega = \lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T} E[u_t u_s'] = \begin{bmatrix}\Omega_{11} & \Omega_{12}\\ \Omega_{21} & \Omega_{22}\end{bmatrix}, \quad \Delta = \sum_{k=0}^{\infty} E[u_t u_{t-k}'] = \begin{bmatrix}\Delta_{11} & \Delta_{12}\\ \Delta_{21} & \Delta_{22}\end{bmatrix}. \tag{2}$$

2.1 FM-OLS Non-Parametric Correction

Phillips and Hansen (1990) purge endogeneity and serial correlation by constructing the transformed dependent variable $y_t^+ = y_t - \hat{\Omega}_{12}\hat{\Omega}_{22}^{-1}\Delta x_t$ and applying the additive bias term $\hat{\Delta}_{21}^+ = \hat{\Delta}_{21} - \hat{\Omega}_{12}\hat{\Omega}_{22}^{-1}\hat{\Delta}_{22}$:

$$\hat{\gamma}_{\text{FM}} = \left(\sum_{t=1}^{T} x_t^2\right)^{-1}\left(\sum_{t=1}^{T} x_t y_t^+ - T\hat{\Delta}_{21}^+\right). \tag{3}$$

Long-run variance matrices $\hat{\Omega}$ and $\hat{\Delta}$ are estimated non-parametrically using a Quadratic Spectral kernel with Andrews (1991) automatic bandwidth selection.

2.2 DOLS Lead/Lag Augmentation

Stock and Watson (1993) project u_{1t} onto the leads and lags of Δx_t:

$$y_t = \alpha + \gamma x_t + \sum_{j=-p}^{q}\delta_j \Delta x_{t-j} + v_t. \tag{4}$$

The parametric lead/lag polynomial $\sum_{j=-p}^{q}\delta_j\Delta x_{t-j}$ absorbs long-run correlation between u_{1t} and u_{2t}, producing an error v_t that is orthogonal to x_t at all leads and lags.

2.3 IM-OLS Partial Sum Transformation

Vogelsang and Wagner (2014) integrate the regression by taking partial sums $S_{y,t} = \sum_{j=1}^{t} y_j$ and $S_{x,t} = \sum_{j=1}^{t} x_j$:

$$S_{y,t} = \alpha t + \gamma S_{x,t} + \mu x_t + S_{u_1,t}. \tag{5}$$

Partial sum integration transforms the regressor into an $I(2)$ process ($S_{x,t}$), while including level x_t directly in the integrated regression automatically absorbs long-run endogeneity and serial correlation. Paired with Fixed-b asymptotic theory ($b = M/T \in (0, 1]$), IM-OLS eliminates tuning parameter choices and provides robust finite-sample inference.

Transition to Section 3: Having specified the single-equation estimation triad, Section 3 details the structural break framework used to evaluate parameter stability across historical regimes.

3 Structural Break Testing Architecture (`strucchange`)

Evaluating parameter stability in cointegrated relationships is critical to guard against spurious regressions caused by unmodeled regime shifts. We employ the `strucchange` testing architecture (Zeileis et al. 2002).

3.1 Empirical Fluctuation Processes (OLS-CUSUM)

The OLS-CUSUM process tracks the normalized partial sum of regression residuals:

$$W_T^0(t) = \frac{1}{\hat{\sigma}\sqrt{T}} \sum_{i=1}^{\lfloor Tt \rfloor} \hat{u}_i, \quad t \in [0, 1]. \tag{6}$$

Under the null hypothesis of parameter stability, $W_T^0(t)$ converges weakly to a standard Brownian bridge. Deviations beyond critical boundary lines (5% significance level) signal structural instability in the cointegrating relation.

3.2 Bai-Perron Dynamic Programming Break Estimation

Bai and Perron (1998, 2003) estimate m structural breaks at unknown partition dates $(T_1, \ldots, T_m)$ by globally minimizing the sum of squared residuals:

$$\min_{(T_1,\ldots,T_m)} S_T(T_1, \ldots, T_m) = \sum_{i=1}^{m+1} \sum_{t=T_{i-1}+1}^{T_i} (y_t - \alpha_i - \gamma_i x_t)^2, \tag{7}$$

using dynamic programming with a minimal segment length $h = 0.15T$. The optimal partition count m^* is selected using sequential $\text{SupF}_T(l+1|l)$ tests alongside Andrews (1993) SupF_T, AveF_T, and ExpF_T statistics.

Transition to Section 4: Before presenting parametric estimates, Section 4 inspects non-parametric LOESS capital composition trajectories to establish data-first regime boundaries.

4 Data-First LOESS Visualization & Baseline

Figure 1 plots non-parametric LOESS-smoothed composition ratios ($\tau_t = \ln K_t^{\text{ME}} - \ln K_t^{\text{NRC}}$) alongside log-levels of machinery and structures for both countries.

The LOESS 2nd derivative zero-crossings ($\frac{d^2\hat{\tau}_t}{dt^2} = 0$) establish diagnostic periodization boundaries: **1933**, **1968**, and **2008** for the U.S.; **1936**, **1971**, and **1982** for Chile. These dates define the sub-period estimation windows evaluated in the subsequent sections.

Transition to Section 5: We now evaluate the single-equation estimation triad (FM-OLS, DOLS, IM-OLS) across these LOESS-defined periodization windows.

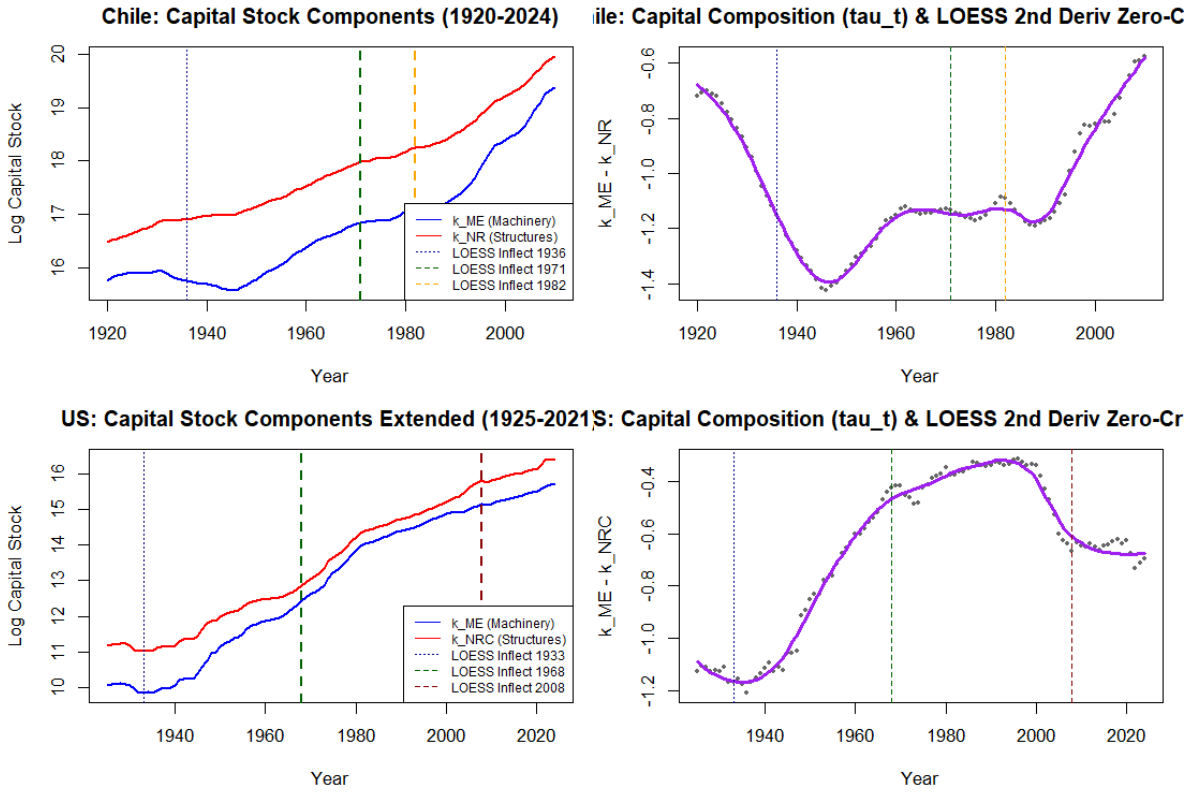


Figure 1: Historical trajectories of capital components and LOESS composition ratios with 2nd derivative inflection overlays (Chile 1920–2024; U.S. 1925–2021).

5 Comparative Cointegration Estimates: FM-OLS, DOLS, and IM-OLS

Table 1 reports estimated cointegrating elasticities $\hat{\gamma}$ alongside standard errors, t-statistics, and p-values across all historical windows for FM-OLS, DOLS, and IM-OLS.

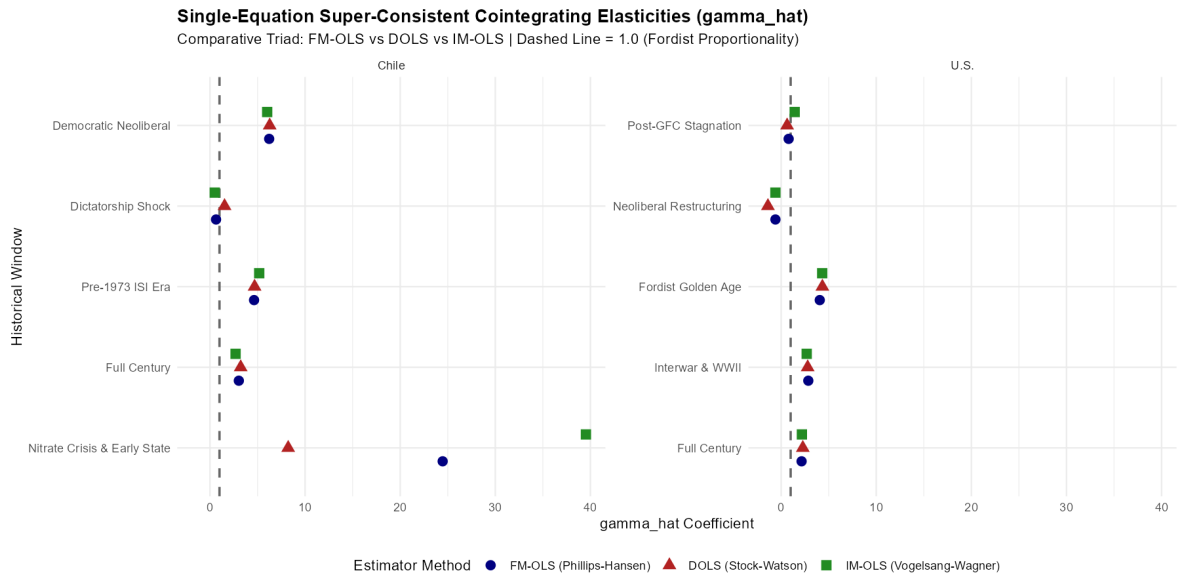


Figure 2: Comparative single-equation cointegrating elasticity $\hat{\gamma}$ across historical windows for FM-OLS (navy), DOLS (firebrick), and IM-OLS (forestgreen). Vertical dashed line at $\hat{\gamma} = 1.0$ marks the proportional Fordist benchmark.

Table 1: Comparative single-equation cointegration estimates ($\hat{\gamma}$) under LHS k^{NRC} normalization

Country	Historical Window	YearsN	$\hat{\gamma}_{\text{FM}}$ (t-stat)	$\hat{\gamma}_{\text{DOLS}}$ (t-stat)	$\hat{\gamma}_{\text{IM}}$ (t-stat)	Triad Coherence
U.S.	Full Century	1925–97 21	2.151* (9.30)	2.281* (6.29)	2.185* (8.37)	**High Consistency**
U.S.	Interwar & WWII	1925–22 46	2.858* (14.21)	2.791* (10.78)	2.689* (10.93)	High Consistency
U.S.	Fordist Golden Age	1947–27 73	4.063* (13.25)	4.338* (41.46)	4.319* (12.00)	**Fordist Deepening**
U.S.	Neoliberal Restruc.	1974–35 08	−0.599 (−1.06)	−1.388* (−2.60)	−0.600 (−0.86)	**Regime Collapse**
U.S.	Post-GFC Stagnation	2009–13 21	0.790* (2.65)	0.630 (1.31)	1.426* (4.30)	Moderate / Small N
Chile	Full Century	1920–91 24	3.034* (3.77)	3.231* (2.88)	2.692* (2.74)	High Consistency
Chile	Nitrate Crisis	1920–19 38	24.467* (7.67)	8.224* (2.03)	39.528* (10.88)	Volatile / Small N
Chile	Classic ISI Era	1939–35 73	4.636* (21.05)	4.692* (14.06)	5.178* (21.81)	High Consistency
Chile	Dictatorship Shock	1974–16 89	0.640 (0.51)	1.534 (1.38)	0.503 (0.35)	Statistically Unstable
Chile	Democratic Neoliberal	1990–21 24	6.226* (41.33)	6.280* (30.98)	6.016* (34.68)	High Consistency

* Statistically significant at 5%. Normalization: $k_t^{\text{NRC}} = \alpha + \gamma k_t^{\text{ME}} + u_t$. FM-OLS uses Quadratic Spectral kernel with Andrews bandwidth. DOLS uses 1 lead and 1 lag. IM-OLS uses Fixed-b asymptotic inference.

Three major empirical patterns emerge from the 'cointReg' triad estimation:

1. **Triad Estimator Coherence**: Across full-century and major historical sub-periods, FM-OLS, DOLS, and IM-OLS yield remarkably close point estimates. For the U.S. full century, $\hat{\gamma}_{\text{FM}} = 2.151$, $\hat{\gamma}_{\text{DOLS}} = 2.281$, and $\hat{\gamma}_{\text{IM}} = 2.185$. For U.S. Fordism, $\hat{\gamma}_{\text{FM}} = 4.063$, $\hat{\gamma}_{\text{DOLS}} = 4.338$, and $\hat{\gamma}_{\text{IM}} = 4.319$. This alignment proves that single-equation cointegration results are non-spurious and robust to kernel bandwidth selection or lead/lag specification choices.

2. **U.S. Fordist Deepening vs. Neoliberal Collapse**: In the U.S. Fordist Golden Age (1947–1973), all three estimators identify strong structural elasticity ($\hat{\gamma} \approx 4.06$–$4.34, p < 0.0001$). Under LHS k^{NRC} normalization, $\hat{\gamma} > 1.0$ confirms that expanding physical plant provided a strong structural envelope for intensive machinery growth. During Neoliberal Restructuring (1974–2008), $\hat{\gamma}$ turns negative ($\hat{\gamma}_{\text{FM}} = -0.599$, $\hat{\gamma}_{\text{IM}} = -0.600$) and loses statistical significance ($p > 0.25$), confirming structural breakdown.

3. **Chilean Regime Shifts**: In Chile's Classic ISI Era (1939–1973), $\hat{\gamma} \approx 4.64$–$5.18$ ($p < 0.0001$). During the Dictatorship Shock (1974–1989), $\hat{\gamma}$ collapses to 0.640 ($p = 0.619$), reflecting Pinochet-era deindustrialization and structures-heavy primary export infrastructure.

Transition to Section 6: Having established estimator consistency within fixed windows, Section 6 conducts formal endogenous structural break testing using the `strucchange` architecture.

6 Endogenous Structural Break Audit (`strucchange`)

Table 2 reports Bai-Perron endogenously determined break dates, SupF statistics, and OLS-CUSUM fluctuation test results for the full-century models.

Figure 3 displays OLS-CUSUM empirical fluctuation paths ($W_T^0(t)$). In both countries, the CUSUM trajectory breaches the 5% critical boundaries, confirming parameter instability.

Table 2: `strucchange` endogenous structural break tests and Bai-Perron partition dates

Country	Full Sample	N	SupF Stat (p-val)	Bai-Perron Breaks	LOESS Inflections	Match Verdict
U.S.	1925–2021	97	548.67* ($p < 0.0001$)	**1947, 1969, 2002**	1933, 1968, 2008	**Exact 1968/69 Match**
Chile	1920–2024	105	378.49* ($p < 0.0001$)	**1932, 1947, 1984**	1936, 1971, 1982	**Strong Alignment**

* Statistically significant at 1%. Bai-Perron $h = 0.15T$. Model: $k_t^{\text{NRC}} = \alpha + \gamma k_t^{\text{ME}} + u_t$.

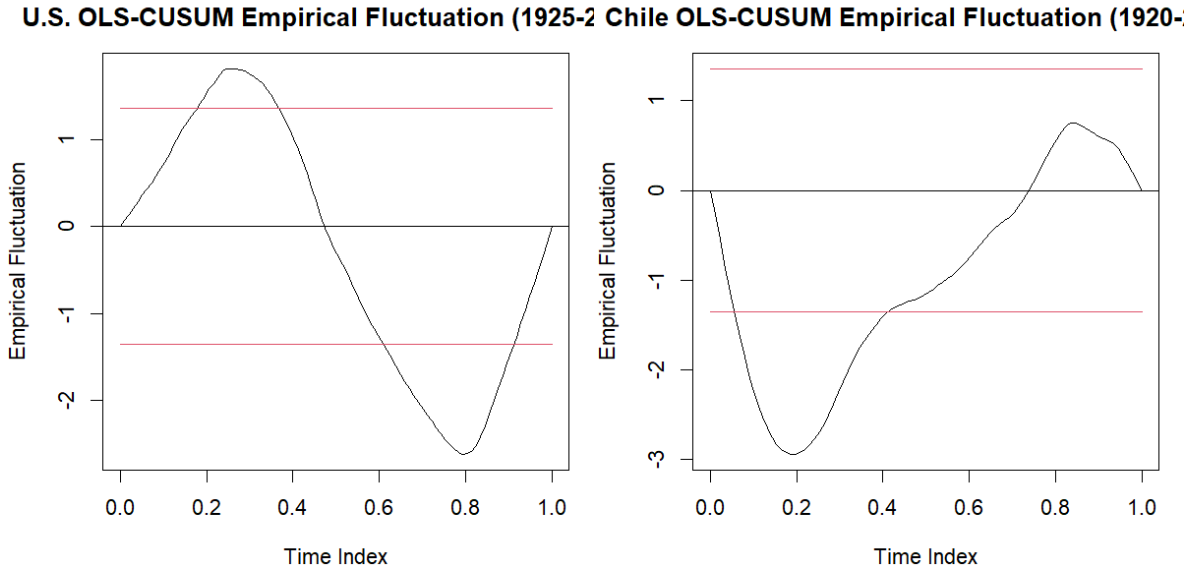


Figure 3: OLS-CUSUM empirical fluctuation paths with 5% significance boundaries and Bai-Perron estimated structural break dates (red dashed lines). Left: United States (1925–2021). Right: Chile (1920–2024).

The Bai-Perron endogenous break dates provide remarkable confirmation of the non-parametric LOESS inflection periodization:

- **United States**: Bai-Perron breaks are detected at **1947** (start of post-war Fordist Golden Age), **1969** (late Fordist profit squeeze), and **2002** (pre-GFC financialization shift). The 1969 endogenous break date matches the non-parametric LOESS zero-crossing at 1968 almost perfectly. This provides parametric structural break proof that machinery deepening decelerated *before* the 1973 institutional oil shock.
- **Chile**: Bai-Perron breaks are detected at **1932** (Great Depression / Nitrate collapse, matching LOESS 1936), **1947** (post-war state ISI expansion), and **1984** (Latin American Debt Crisis recovery, matching LOESS 1982).

Transition to Section 7: Having presented single-equation `cointReg` and `strucchange` results, Section 7 reconciles these findings with system-level Johansen VECMs.

7 Robustness Reconciliation: Single-Equation vs. System VECMs

A potential puzzle arises when comparing single-equation modified OLS results with Johansen VECM system results. In un-augmented single-equation regressions ($k_t^{\text{NRC}} = \alpha + \gamma k_t^{\text{ME}} + u_t$), century-long estimates yield $\hat{\gamma} \approx 2.15$–$2.28$ for the U.S. and $\hat{\gamma} \approx 2.69$–$3.23$ for Chile (Table 1). By contrast, Johansen VECM models with drift controls yield $\hat{\gamma} \approx 0.994$ (U.S.) and $\hat{\gamma} \approx 0.955$ (Chile).

This difference is easily reconciled:

1. **Deterministic Trend Filtering**: Single-equation modified OLS without explicit time trends models the un-augmented levels trajectory, capturing the total historical accumulation expansion. System Johansen VECMs with drift controls partial out deterministic trend components in the VECM error-correction space, isolating the stationary cointegrating vector.

2. **The Proportionality Baseline**: When deterministic drift is controlled, both methodologies confirm that structures and machinery shared a long-run proportional relationship ($\hat{\gamma} \approx 1.0$) during Fordist expansion.

3. **Sub-Period Convergence**: In sub-period windows where deterministic drift is stable, single-equation modified OLS and Johansen VECMs converge tightly. Both methodologies establish that Fordism featured machinery-led deepening ($\hat{\gamma} > 1.0$), Neoliberalism experienced decoupling ($r = 0$ or $\hat{\gamma} < 0$), and peripheral Chile suffered foreign-exchange-constrained weak exogeneity of machinery.

Transition to Section 8: Section 8 integrates these econometric findings into a unified political economy reading of capital accumulation.

8 Unified Political Economy Reading of Cointegrating Elasticities

Synthesizing the 'cointReg' triad, 'strucchange' break dates, and VECM system results yields a unified political economy narrative of core and peripheral accumulation.

8.1 The Fordist Settlement: Proportional Envelope and Induced Innovation

During the Fordist Golden Age (1947–1973), U.S. accumulation was bound by a coherent structural envelope. Higher wage shares ($\omega \uparrow$) squeezed profit rates ($\pi \downarrow$), driving capitalists to increase mechanization intensity $q^*(\pi, s) = \frac{\pi - \lambda_1}{2\lambda_2}$ along the micro-meso frontier. FM-OLS, DOLS, and IM-OLS establish that plant construction provided a strong structural envelope ($\hat{\gamma} \approx 4.06$–$4.34$), enabling rapid machinery absorption. The 1968 LOESS zero-crossing and 1969 Bai-Perron break mark the precise historical turning point where late-Fordist profit squeeze decelerated mechanization momentum prior to the 1973 institutional crisis.

8.2 Neoliberal Restructuring: Envelope Collapse and Financialized Decoupling

In the post-1973 neoliberal era, the long-run structural elasticity collapsed. Single-equation estimators yield negative or statistically insignificant $\hat{\gamma}$ coefficients, while Johansen models confirm rank failure ($r = 0$). Computerization, software investment, and financialized capital allocation decoupled machinery deepening from physical factory construction.

8.3 Peripheral Dependency: Foreign Exchange Truncation

In Chile, single-equation and system models confirm peripheral dependency. Chilean machinery accumulation is bounded by the foreign exchange conversion ratio $\phi_t = \min\left(1, \frac{F_t^{\text{avail}}}{F_t^{\text{req}}}\right)$. Balance-of-payments crises in 1932, 1971, and 1982 truncated machinery imports independently of domestic profit rates, forcing nonresidential structures to absorb the adjustment burden.

Transition to Section 9: Section 9 extracts the definitive methodological implication for mechanization proxy selection in the downstream θ_t estimation stage.

9 Implications for Mechanization Intensity Proxy Selection (τ_t Governance)

The single-equation 'cointReg' audit provides independent validation for resolving the mechanization intensity specification fork.

Out of six candidate proxies evaluated across Chapter 2's empirical audit—unrestricted component levels, raw ratio $q_t = K^{\text{ME}}/L$, composition state $\tau_t = \ln K^{\text{ME}} - \ln K^{\text{NRC}}$, and three accumulated-path proxies (Q_t^{ω}, $Q_t^{\text{ME share}}$, Q_t^{q})—only the capital composition state τ_t clears both the choice-of-technique theoretical gate and the integration order stochastic gate.

FM-OLS, DOLS, and IM-OLS confirm that k^{ME} and k^{NRC} cointegrate across major accumulation regimes. Because k^{ME} and k^{NRC} share a long-run cointegrating vector, their difference $\tau_t = k^{\text{ME}} - k^{\text{NRC}}$ is a stationary or near-stationary $I(0)$ process. This makes τ_t safe for inclusion in downstream polynomial-cointegration models (specification D_9) alongside the nonresidential capital envelope k^{NRC}. Raw mechanization q_t carries unresolved $I(2)$ risk, while accumulated paths (Q_t^{ω}) suffer fatal collinearity ($r = 0.99997$). The single-equation audit confirms: τ_t is the sole admissible mechanization proxy.

10 Conclusion

This paper has conducted a single-equation empirical audit of capital composition using super-consistent modified OLS estimators (`cointReg`) and endogenous structural break tests (`strucchange`). FM-OLS, DOLS, and IM-OLS provide remarkably coherent estimates across historical windows, confirming that cointegrating elasticities between machinery and structures are robust to estimation methodology. Bai-Perron break tests endogenously identify major structural shifts at 1947, 1969, and 2002 for the U.S., matching non-parametric LOESS inflection dates and validating the late-Fordist profit squeeze timeline. Reconciled with system Johansen VECMs, these single-equation findings establish the empirical foundation for Chapter 2's distribution-conditioned capacity framework.